

To the Editorial Board

Journal of Statistical Mechanics: Theory and Experiment

Dear Editors,

Please consider the enclosed manuscript, “Density-gated noise rectification in a Vicsek–Couzin flock: one dense-phase mechanism, additive at matched decorrelation,” for publication in JSTAT as an original research article. We suggest the section *Classical statistical mechanics, equilibrium and non-equilibrium*; the work also touches *Interdisciplinary statistical mechanics* through its motivation from collective motion in living systems.

The paper makes a methodological point, and makes it by way of a concrete model. Flocking studies routinely compare noise distributions of different shape at matched nominal amplitude η , but that comparison is not neutral: a wrapped α -stable kick of scale η decorrelates a heading at the rate $\Gamma = 1 - \exp(-\eta^\alpha)$, which depends on the tail index as strongly as on the amplitude. Varying the shape at fixed η therefore varies the effective noise as well, and we show that this confound is large enough to manufacture an apparent cooperative effect that survives several independent consistency checks before a matched-decorrelation control dissolves it.

The vehicle is a Vicsek–Couzin flock in which one sigmoid in the local neighbour count gates both the self-propulsion speed and the stability index of the angular noise, making a crowded particle at once slow and Gaussian-like and an isolated one fast and Cauchy-like. That is deliberately the arrangement most favourable to a synergy, which is what makes a negative answer informative. A four-way ablation on a single code path, over nine sizes up to $L = 256$ at ten seeds, separates the channels: heavy-tailed reorientations alone grow no size-dependent susceptibility, only the motility channel builds a phase-separated dense phase, and the noise-shape channel acts exclusively inside it. Compared at each mode’s own near-critical noise the full model appears to gain a super-additive boost, $\Delta_9 = +0.45$; under the matched-decorrelation control it falls to $+0.02$, additive within error, while the dense static and dynamic correlation gaps reverse sign. The coupling reduces to a density-gated lowering of the dense-phase effective noise, governed by the motility threshold alone and contingent on a metric kernel.

The components being individually well studied, we have positioned the contribution narrowly. Density-dependent motility with alignment is established, and a density-dependent noise *amplitude* is not new either; what has not been done is to make the *shape* of the turning law a function of local density and couple it to a density-dependent speed, a cell the framework of Clusella and Pastor-Satorras admits and then sets aside as beyond its scope. In the same spirit we report the finite-size slopes as effective growth rates rather than asymptotic exponents, noting that our full-mode value $+1.49$ is indistinguishable from the $\gamma/\nu = 1.45(2)$ established for the standard Vicsek model while the single-channel ablation excludes it, and that the susceptibility peak still migrates on our window. The internal falsifications are reported rather than buried.

All code and data are openly available at <https://github.com/yayayou47/two-feedback-vicsek> under the MIT licence, and every figure reproduces bit-for-bit from a single seeded generator. The manuscript is original, is not under consideration elsewhere, and its two authors declare no competing interests; one supplementary-material PDF accompanies it.

Thank you for considering our submission.

Sincerely,



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